

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 4 / 12

Fisher Information: Two Routes to the Same Answer

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Contents

1	Fisher Information for $\mathcal{N}(\mu, 1)$	2
1.1	Definition and operator view	2
1.2	Two routes to $\mathcal{I}$	3
1.3	Score surface and the pre-integrand $(x - \mu)^2$	3
1.4	Forming the pre-integral curve	4
1.5	Computing $\mathcal{I}_{\mathcal{N}(\mu, 1)}(\mu)$ via the generic route	5
1.6	Regular route: the score surface and its derivative	5
1.7	Sections of the flat plane	6
1.8	Product, negation, and the final result	7
1.9	Existence of Fisher information and the information identity	8

1 Fisher Information for $\mathcal{N}(\mu, 1)$

1.1 Definition and operator view

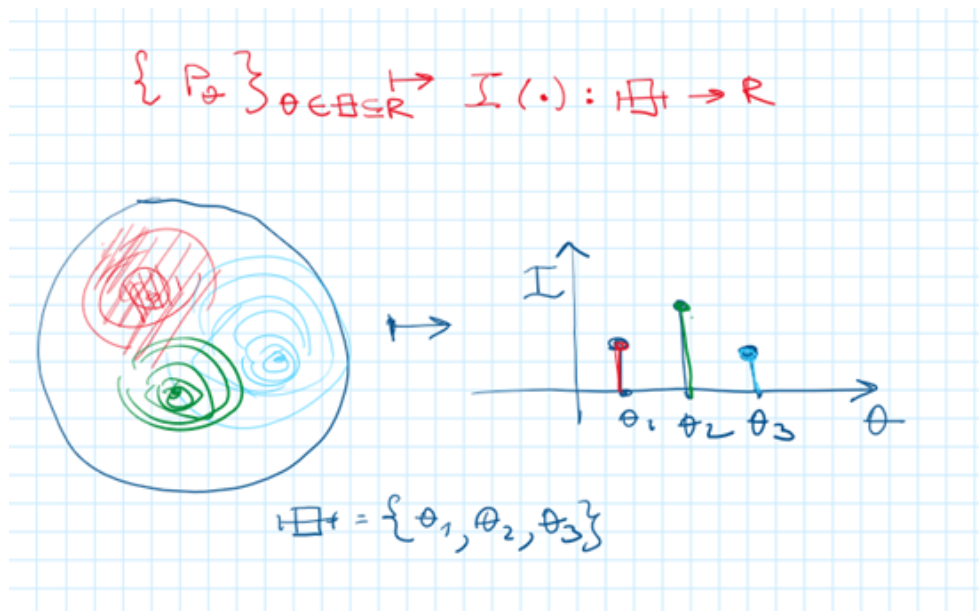
The *Fisher information* is defined as the variance of the score:

$$\mathcal{I}(\theta) = \text{Var}_\theta[s(X, \theta)] = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right] = \int_{\mathbb{R}} \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right)^2 f(x; \theta) dx.$$

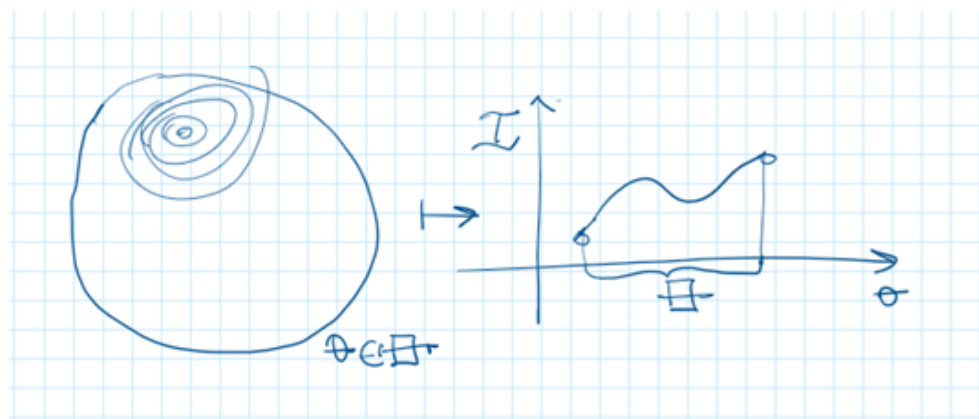
The last equality holds in the absolutely continuous case (Lebesgue density exists). Since it is an expectation of a square, $\mathcal{I}(\theta) \geq 0$ always.

Structurally, $\mathcal{I}$ is an operator on one-parametric families of measures: each family $\{P_\theta\}_{\theta \in \Theta}$ is mapped to a univariate function $\mathcal{I}(\cdot): \Theta \rightarrow [0, \infty)$. The parameter space Θ must be a continuum because we need to differentiate in θ .

The picture below shows the toy case $\Theta = \{\theta_1, \theta_2, \theta_3\}$: a family of measures on the left maps to three scalar values on the right.



When Θ is a continuum the output is a curve over θ :



1.2 Two routes to $\mathcal{I}$

In the **generic** (absolutely continuous) case the formula involves two operations: (1) square the score, (2) integrate against f_θ :

$$\mathcal{I}(\theta) = \int_{\mathbb{R}} s^2(x, \theta) f_\theta(x) dx.$$

Under **regularity conditions** (differentiation under the integral sign is valid) a third equivalent formula holds:

$$\mathcal{I}(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \ln L_x(\theta) \right] = -\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} s_x(\theta) \right].$$

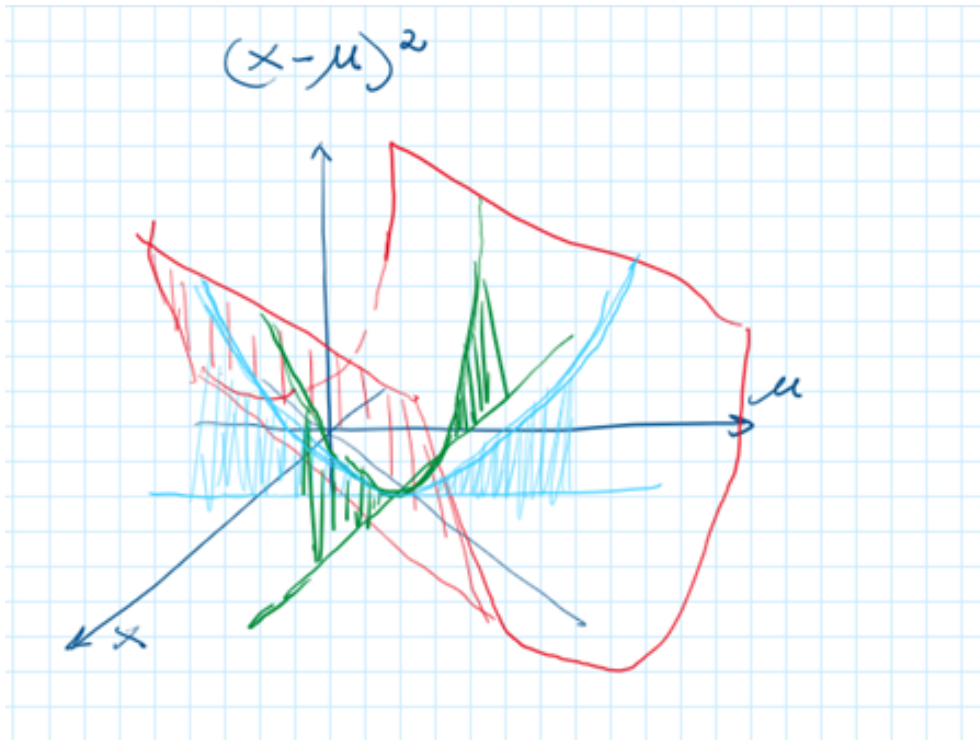
This uses three operations: (1) differentiate the score in θ , (2) integrate against f_θ , (3) negate. We work out both for $\mathcal{N}(\mu, 1)$.

1.3 Score surface and the pre-integrand $(x - \mu)^2$

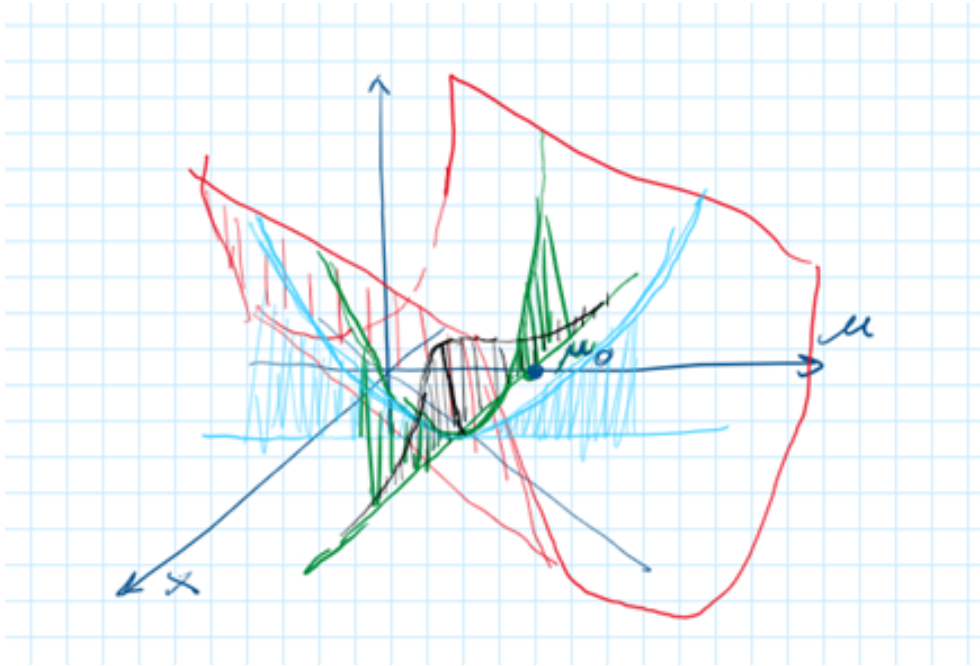
For the location family $f_\mu(x) = \varphi(x - \mu)$:

$$\ell_x(\mu) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2}(x - \mu)^2, \quad s_x(\mu) = \frac{\partial \ell_x}{\partial \mu} = x - \mu.$$

The pre-integrand for the generic formula is $(x - \mu)^2$ — a perfect square. The bivariate surface $(x - \mu)^2$ over the (x, μ) -plane:

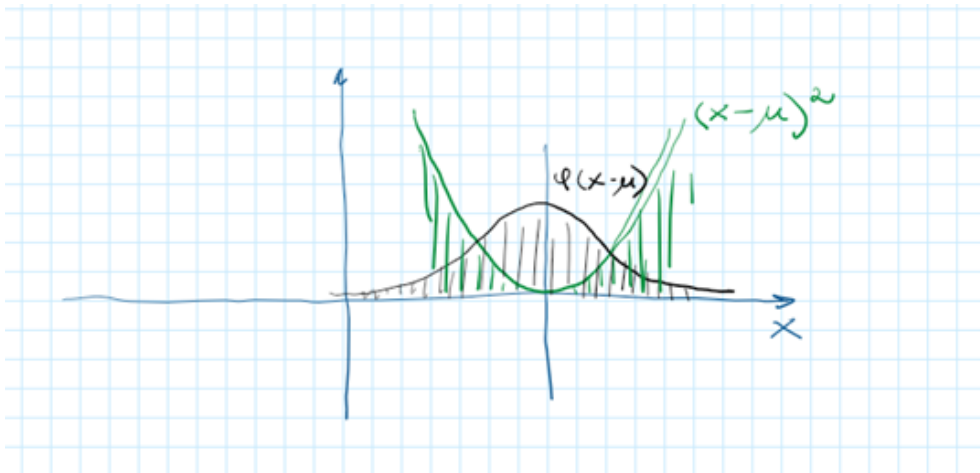


Fix $\mu = \mu_0$. The green density $\varphi(x - \mu_0)$ is added as a section:

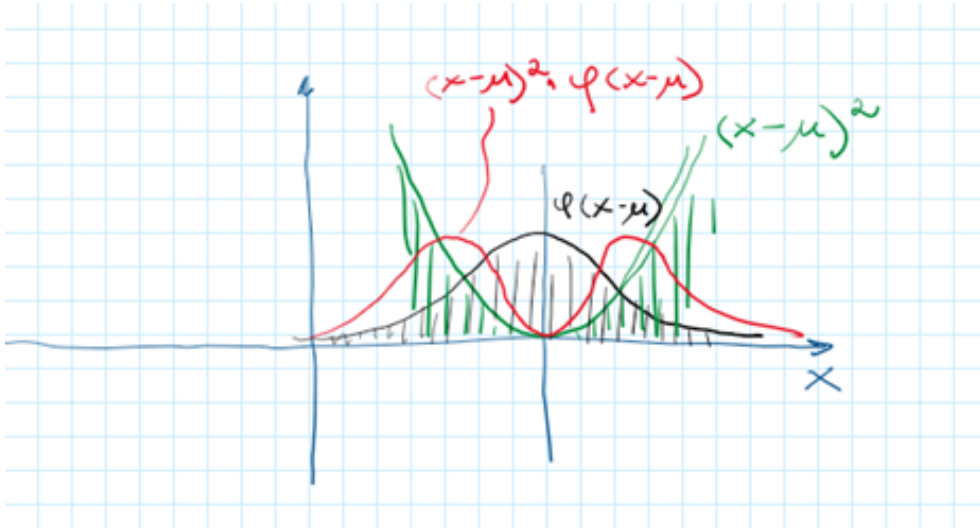


1.4 Forming the pre-integral curve

At fixed μ_0 , the two factors in the integrand are $(x - \mu_0)^2$ (green, a parabola) and $\varphi(x - \mu_0)$ (black, a bell curve):



Their product $(x - \mu_0)^2 \cdot \varphi(x - \mu_0)$ is the pre-integral curve (red). It is non-negative and Lebesgue integrable; its integral is $\mathcal{I}(\mu_0)$:



1.5 Computing $\mathcal{I}_{\mathcal{N}(\mu,1)}(\mu)$ via the generic route

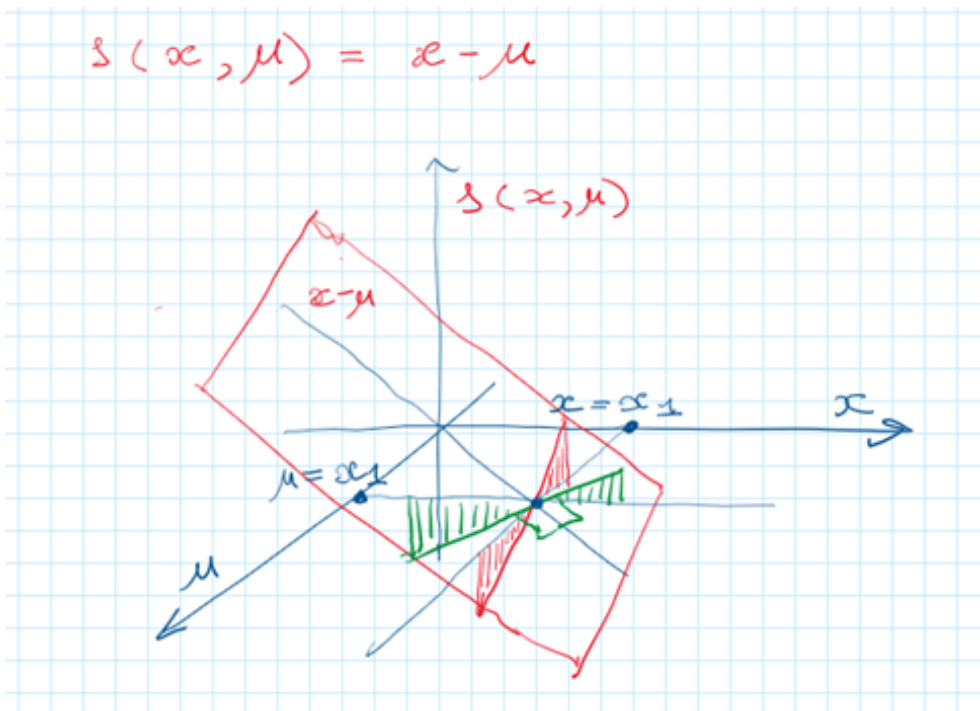
Integrating with the substitution $y = x - \mu$:

$$\mathcal{I}_{\mathcal{N}(\mu,1)}(\mu) = \int_{-\infty}^{\infty} (x - \mu)^2 \varphi(x - \mu) dx = \int_{-\infty}^{\infty} y^2 \varphi(y) dy = \text{Var}(Z) = 1, \quad Z \sim \mathcal{N}(0, 1).$$

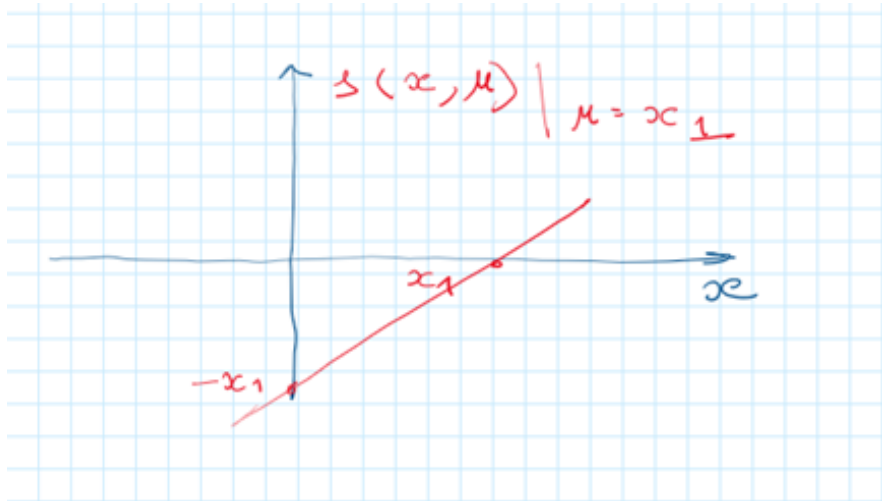
The result does not depend on μ : $\mathcal{I} \equiv 1$ for this family. In general $\mathcal{I}(\theta)$ depends on θ .

1.6 Regular route: the score surface and its derivative

The bivariate surface $s(x, \mu) = x - \mu$ over the (x, μ) -plane, with two sections highlighted:



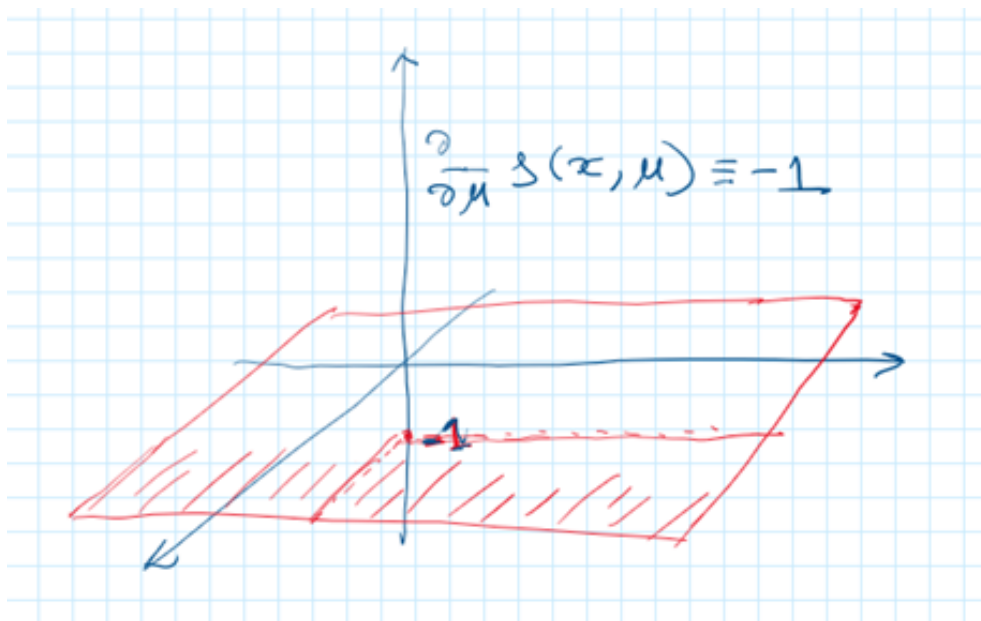
Section at fixed $\mu = x_1$ (score as a function of x) — a line of slope +1 crossing zero at $x = x_1$:



Now differentiate the score in μ :

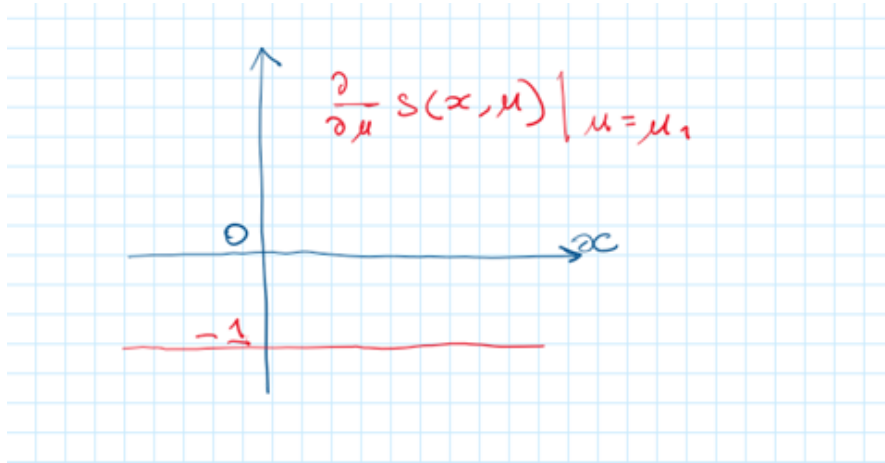
$$\frac{\partial}{\partial \mu} s(x, \mu) = \frac{\partial}{\partial \mu} (x - \mu) = -1.$$

Already at this level the result does not depend on x . The bivariate surface $\frac{\partial}{\partial \mu} s(x, \mu) \equiv -1$ is a flat horizontal plane at height -1 :

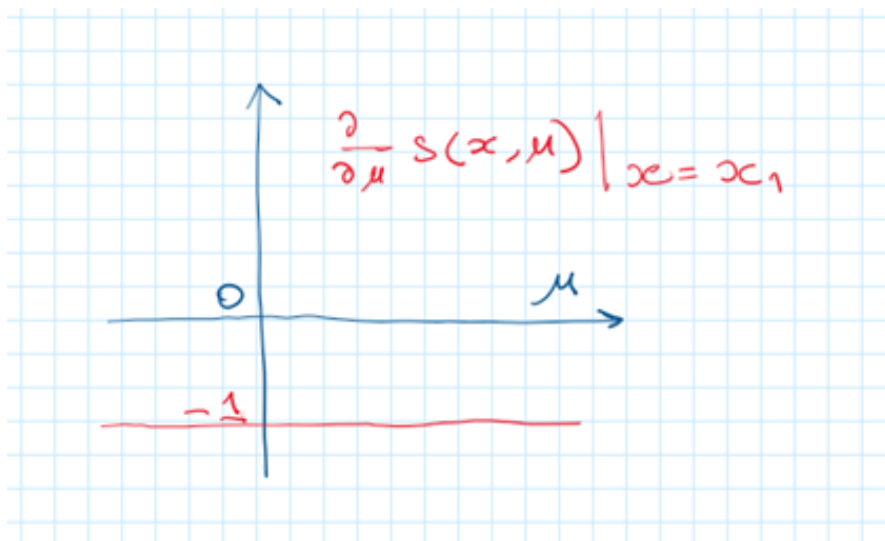


1.7 Sections of the flat plane

Section at fixed $\mu = \mu_1$ (as a function of x) — a horizontal line at -1 , constant in x :

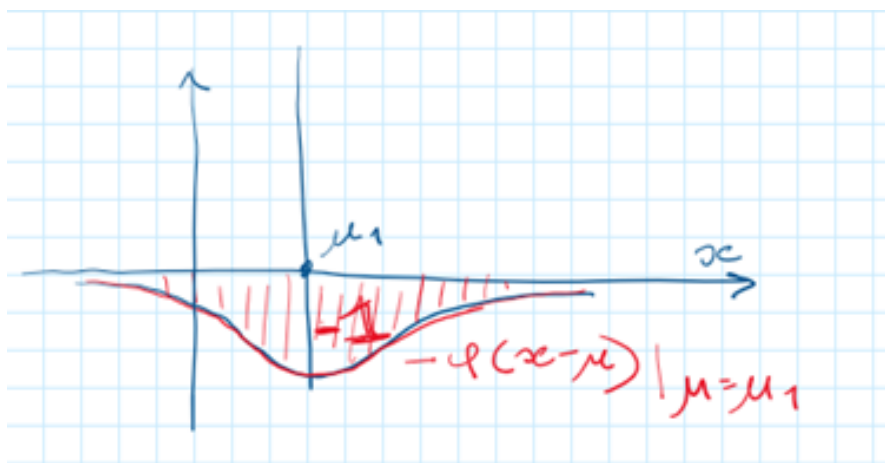


Section at fixed $x = x_1$ (as a function of μ) — also a horizontal line at -1 , constant in μ :

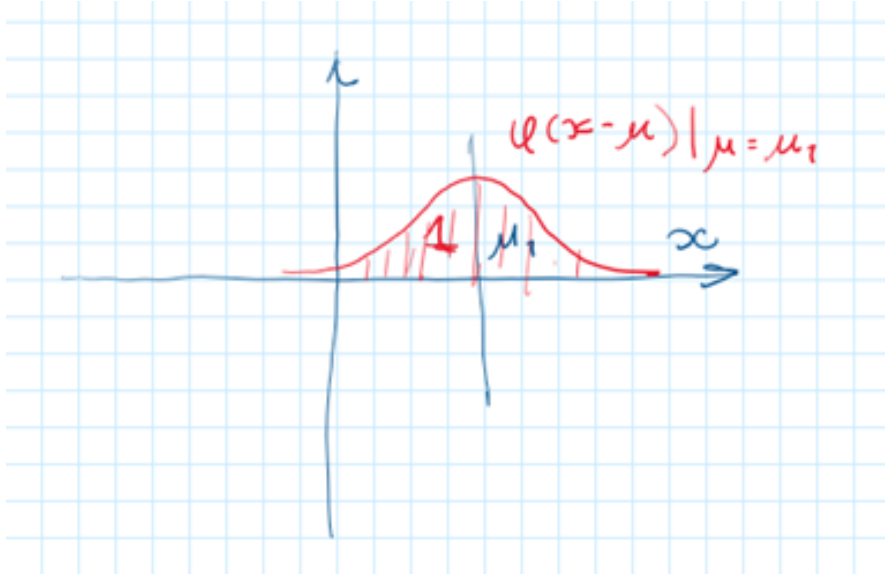


1.8 Product, negation, and the final result

The product $\frac{\partial}{\partial \mu} s(x, \mu) \cdot f_\mu(x) = (-1) \cdot \varphi(x - \mu) = -\varphi(x - \mu)$. At $\mu = \mu_1$ this is the negative density — an upside-down bell:



Negating gives $\varphi(x - \mu)$, a standard bell curve centred at μ_1 , which integrates to 1:



$$\mathcal{I}(\mu) = - \int_{-\infty}^{\infty} \frac{\partial}{\partial \mu} s(x, \mu) \cdot \varphi(x - \mu) dx = \int_{-\infty}^{\infty} \varphi(x - \mu) dx = 1.$$

Both routes confirm $\mathcal{I}_{\mathcal{N}(\mu, 1)} \equiv 1$.

Summary.

Route	Operations	Result
Generic	square s , integrate	$\int y^2 \varphi(y) dy = 1$
Regular	differentiate s , integrate, negate	$-\int (-1) \varphi(x - \mu) dx = 1$

1.9 Existence of Fisher information and the information identity

Existence conditions

The definition $\mathcal{I}(\theta) = \mathbb{E}_{\theta}[s(X, \theta)^2]$ presupposes three things in sequence:

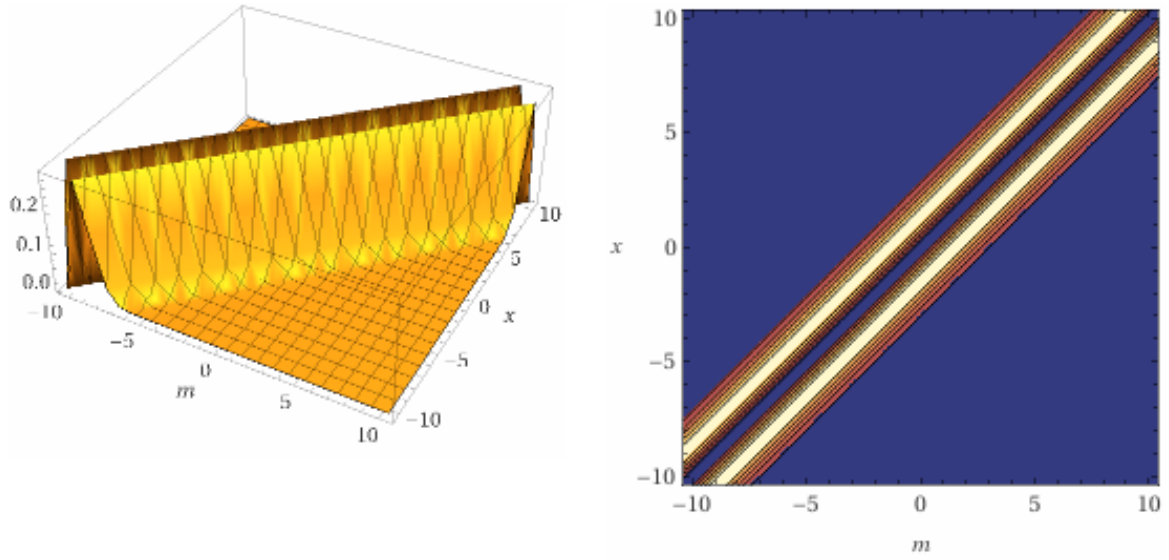
- (1) **Density exists.** The family must be absolutely continuous, so the likelihood $f(x; \theta)$ (Lebesgue density) is well-defined.
- (2) **Score exists.** The density must be differentiable in θ (at least a.e. in x), so that $s(x, \theta) = \partial_{\theta} \ln f(x; \theta)$ makes sense.
- (3) **Score is square-integrable.** The integral $\int s(x, \theta)^2 f(x; \theta) dx$ must be finite. In the worst case it diverges: $\mathcal{I}(\theta) = +\infty$.

Two functions with the same integral

Having worked through both routes we can now state the information identity geometrically. $\mathcal{I}(\theta) = \mathbb{E}_{\theta}[s^2] = -\mathbb{E}_{\theta}[\partial_{\theta} s]$ says that two *different* functions of x have equal integrals against f_{θ} . For $\mathcal{N}(\mu, 1)$ these are

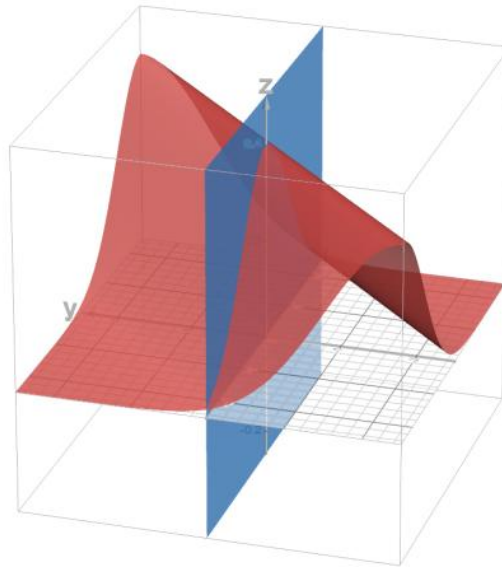
$$g_1(x) = (x - \mu)^2 \varphi(x - \mu), \quad g_2(x) = \varphi(x - \mu),$$

both integrating to 1. They look completely different — g_1 is the two-humped product curve from Section 1.4, g_2 is just the standard bell from Section 1.8 — yet their areas coincide. Left panel: g_1 with φ and $(x - \mu)^2$ shown separately; right panel: $g_2 = \varphi(x - \mu_1)$ at a fixed μ_1 . Wire-frame and top-down views of the full bivariate surface $z = (x - \mu)^2 \varphi(x - \mu)$ are shown below the 2D panels:



Blue-plane intersection: selecting the 2D curve

The surface $z = (x - \mu)^2 \varphi(x - \mu)$ with a vertical blue plane at $\mu = \mu_0$. The intersection of the red surface and the blue plane is exactly $g_1(\cdot; \mu_0)$; its integral equals $\mathcal{I} = 1$:



A second viewing angle of the same configuration:

